

app

Data Upload

Data Profile

Business Dashboard

AI Insights

Ask Your Data

Forecasting

Executive Report



Autonomous Business Intelligence & Decision Support Platform

Transform raw business data into AI-powered insights, intelligent dashboards, predictive forecasts, and executive reports.

Platform Highlights

Interactive Dashboards

- Dynamic KPI Cards
- Interactive Charts
- Business Performance Tracking

AI Business Analysis

- Executive Insights
- Trend Detection
- Business Recommendations

Predictive Analytics

- Forecast Future Trends
- AI Executive Reports
- Decision Support

Why InsightIQ AI?

- ✓ Supports Multiple Business Datasets
- ✓ Automatic Semantic Column Detection
- ✓ Intelligent KPI Generation
- ✓ AI-Powered Business Insights
- ✓ Interactive Business Dashboards
- ✓ Predictive Forecasting
- ✓ Executive Decision Support
- ✓ Minimal Manual Configuration

Explore the Platform

Data Upload

Upload CSV or Excel datasets and let InsightIQ AI automatically understand your business data.

Business Dashboard

View KPIs, interactive visualizations, and business trends.

Forecasting

Predict future business performance using machine learning.

Data Profiling

Explore missing values, statistics, data quality, and dataset structure.

AI Business Analyst

Generate business insights, identify patterns, and receive recommendations.

Executive Report

Generate AI-powered executive summaries for business leaders.

Data Upload

Upload your business dataset.

Supported formats:

-  ZIP
-  CSV
-  Excel (.xlsx)

Choose Dataset



archive (6).zip
177.8MB



Uploaded : archive (6).zip

Extraction completed

1 tables detected



Table Understanding

```
{
  "fact_tables": []
  "dimension_tables": [
    0 : "PS_20174392719_1491204439457_log.csv"
  ]
  "dataset_type": "generic"
}
```



Business Model Created



Business Model

Shape: (6362620, 11)

Columns:

```
[
  0 : "step"
  1 : "type"
  2 : "amount"
  3 : "nameOrig"
  4 : "oldbalanceOrg"
  5 : "newbalanceOrig"
  6 : "nameDest"
  7 : "oldbalanceDest"
  8 : "newbalanceDest"
  9 : "isFraud"
  10 : "isFlaggedFraud"
]
```

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest
0	1	PAYMENT	9839.64	C1231006815	170136	160296.36	M1979787155	
1	1	PAYMENT	1864.28	C1666544295	21249	19384.72	M2044282225	
2	1	TRANSFER	181	C1305486145	181	0	C553264065	
3	1	CASH_OUT	181	C840083671	181	0	C38997010	
4	1	PAYMENT	11668.14	C2048537720	41554	29885.86	M1230701703	



Available Tables

 PS_20174392719_1491204439457_log.csv

Data Profile

Dataset Understanding

```
{
  "type": "Fraud",
  "data_columns": [
    "numeric_columns": [
      0: "step"
      1: "amount"
      2: "oldbalanceOrig"
      3: "newbalanceOrig"
      4: "oldbalanceDest"
      5: "newbalanceDest"
      6: "isfraud"
      7: "isflaggedfraud"
    ]
    "categorical_columns": [
      0: "type"
      1: "nameOrig"
      2: "nameDest"
    ]
  ]
}
```

Semantic Model

```
{
  "data": "step"
  "amount": "amount"
  "fraud": "isfraud"
  "transaction": "type"
}
```

Generated Metrics

```
{
  "Total Transactions": 6362620
  "Fraud Cases": 513
  "Fraud Rate %": 0.13
}
```

Dataset Intelligence

```
{
  "rows": 6362620
  "columns": 11
  "numeric_columns": [
    0: "step"
    1: "amount"
    2: "oldbalanceOrig"
    3: "newbalanceOrig"
    4: "oldbalanceDest"
    5: "newbalanceDest"
    6: "isfraud"
    7: "isflaggedfraud"
  ]
  "categorical_columns": [
    0: "type"
    1: "nameOrig"
    2: "nameDest"
  ]
  "data_columns": [
    "numeric_columns": [
      0: "step"
      1: "amount"
      2: "oldbalanceOrig"
      3: "newbalanceOrig"
      4: "oldbalanceDest"
      5: "newbalanceDest"
    ]
    "categorical_columns": [
      0: "type"
      1: "nameOrig"
      2: "nameDest"
    ]
  ]
  "id_columns": [
    "numeric_columns": [
      0: "step"
      1: "amount"
      2: "oldbalanceOrig"
      3: "newbalanceOrig"
      4: "oldbalanceDest"
      5: "newbalanceDest"
    ]
  ]
  "dimension_columns": [
    0: "type"
  ]
}
```

Dataset loaded successfully

Preview

step	type	amount	nameOrig	oldbalanceOrig	newbalanceOrig	nameDest	oldbalanceDest
0	1	PAYMENT	1626.64	C123306431	170136	360296.36	16197970155
1	1	PAYMENT	1864.38	C166544325	21249	19384.72	1609430225
2	1	TRANSFER	181	C135480145	181	0	C35254905
3	1	CASH_OUT	181	C84083871	181	0	C3897010
4	1	PAYMENT	11668.14	C104337720	11551	22883.38	1612870178

Dataset Profile

Rows	Columns	Missing Values	Duplicates
6362620	11	0	0
Memory (MB)	Numeric Columns	Categorical Columns	Date Columns
706.22	8	3	0

Column Information

Column	Data Type	Missing Values	Unique Values
step	step	0	743
type	type	0	5
amount	amount	0	510900
nameOrig	nameOrig	0	6353307
oldbalanceOrig	oldbalanceOrig	0	1843844
newbalanceOrig	newbalanceOrig	0	2682586
nameDest	nameDest	0	2722962
oldbalanceDest	oldbalanceDest	0	2624697
newbalanceDest	newbalanceDest	0	2554999
isfraud	isfraud	0	2

Dataset Summary

step	type	amount	nameOrig	oldbalanceOrig	newbalanceOrig	nameDest	oldbalanceDest
count	6362620	6362620	6362620	6362620	6362620	6362620	636
unique	None	5	None	6353307	None	None	2722962
top	None	CASH_OUT	None	C20965219	None	None	C12860495
freq	None	2237508	None	3	None	None	113
mean	243.9872	None	178801.9015	None	833883.1041	855113.6596	None
std	142.352	None	903808.2215	None	208245.6173	2924948.302	None
min	1	None	0	None	0	0	None
25%	158	None	13280.57	None	0	0	None
50%	239	None	74871.94	None	14006	0	None
75%	335	None	208723.4775	None	187318.175	149258.41	None

Missing Values

No missing values found.

Business Dashboard

Detected Dataset Type: FRAUD

Dashboard Data Check

Shape: (6362629, 11)

Columns:

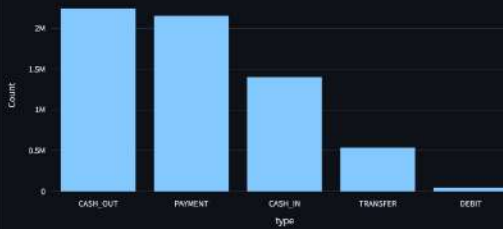
```
[
  0 : "step"
  1 : "type"
  2 : "amount"
  3 : "nameOrig"
  4 : "oldbalanceOrg"
  5 : "newbalanceOrig"
  6 : "nameDest"
  7 : "oldbalanceDest"
  8 : "newbalanceDest"
  9 : "isFraud"
  10 : "isFlaggedFraud"
]
```

Business KPIs

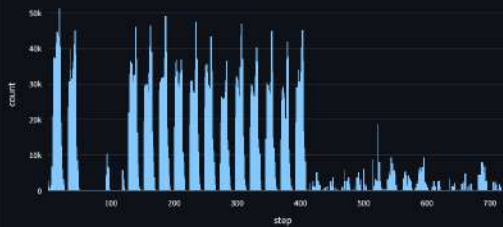
Total Records	Transaction Volume	Avg Transaction Value	Fraud Cases
6362620	1144392...	179861.9	8213
Fraud Rate %	Legit Transactions	Missing Values	
0.13	6354407	0	

Business Visualizations

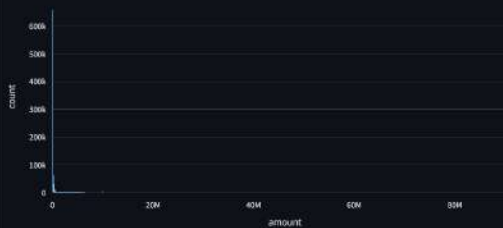
type Distribution



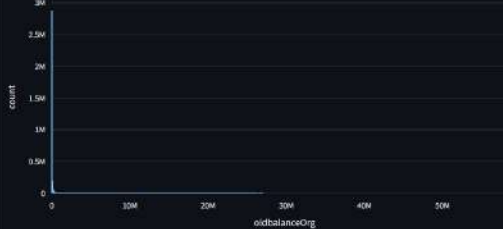
step Distribution



amount Distribution



oldbalanceOrg Distribution



🔍 Ask Your Data

Ask a question about your data

explain the dataset

As an expert Business Intelligence Analyst, I've reviewed your dataset, which appears to contain transactional data with a focus on identifying fraudulent activities. This dataset is a valuable resource for understanding transaction patterns, customer behavior, and ultimately, for developing or enhancing fraud detection systems.

Here's an explanation of each column and some initial business insights:

Dataset Column Explanation & Business Insights

1. step:

- **Explanation:** This column represents a unit of time, likely in hours or days, from the start of the simulation. It doesn't represent specific dates but rather sequential steps in time.
- **Business Insight:** Analyzing transaction volume and fraud incidents across these 'steps' can reveal temporal patterns. For example, are there specific times of day or periods where fraud attempts are more prevalent? This can inform resource allocation for fraud monitoring teams or help in fine-tuning real-time detection rules to be more sensitive during peak risk periods.

2. type:

- **Explanation:** This categorical column describes the type of transaction (e.g., "PAYMENT", "TRANSFER", "CASH_OUT", "CASH_IN", "DEBIT").
- **Business Insight:** This is a critical dimension for fraud analysis. We need to analyze the distribution of fraud across different transaction types. For instance, are "TRANSFER" and "CASH_OUT" transactions disproportionately associated with fraudulent activity compared to "PAYMENT" or "CASH_IN"? Understanding these propensities can help in prioritizing monitoring efforts and developing type-specific fraud detection rules.

3. amount:

- **Explanation:** The value of the transaction.
- **Business Insight:** Analyzing the distribution of transaction amounts for both legitimate and fraudulent transactions is key. Fraudulent transactions might often involve unusually high amounts (to maximize gains) or specific 'round numbers' to evade simple detection rules. Conversely, very small amounts might also be used to test stolen credentials. It's crucial to compare this amount with the changes in `oldBalanceOrig`, `newBalanceOrig`, and `oldBalanceDest` to identify inconsistencies.

4. nameOrig:

- **Explanation:** Unique Identifier for the customer initiating the transaction.
- **Business Insight:** This allows us to track the behavior of originating accounts. Analyzing `nameOrig` can help identify accounts that are frequently involved in suspicious activities, potentially identifying "mule accounts" or compromised customer accounts. Tracking the history of an `nameOrig` can reveal unusual spikes in activity or atypical transaction types.

5. oldBalanceOrig:

- **Explanation:** The balance of the originating customer's account before the transaction.
- **Business Insight:** This is essential for verifying transaction integrity. We expect `oldBalanceOrig - amount` to be approximately `newBalanceOrig` (after accounting for any fees, if applicable, though not present here). Significant discrepancies could indicate data errors or sophisticated fraud attempts where balances are manipulated.

6. newBalanceOrig:

- **Explanation:** The balance of the originating customer's account after the transaction.
- **Business Insight:** Together with `oldBalanceOrig` and `amount`, this allows for a crucial check: Does `oldBalanceOrig - amount = newBalanceOrig`? For fraudulent transactions, especially "TRANSFER" or "CASH_OUT", it's common to see `newBalanceOrig` become 0, indicating an attempt to drain the account completely.

7. nameDest:

- **Explanation:** Unique Identifier for the customer or merchant receiving the transaction.
- **Business Insight:** Similar to `nameOrig`, tracking `nameDest` can help identify destination accounts that are frequently used to receive fraudulent funds. This could point to known fraudsters, money laundering operations, or compromised merchant accounts. For "PAYMENT" transactions, `nameDest` usually represents a merchant, while for "TRANSFER"/"CASH_OUT", it could be another customer.

8. oldBalanceDest:

- **Explanation:** The balance of the destination account before the transaction.
- **Business Insight:** For certain types of fraud, particularly "TRANSFER" or "CASH_OUT" where money is moved to a fraudster's account, `oldBalanceDest` might often be 0, implying a newly opened or "mule" account created solely for receiving illicit funds. Analyzing this for different transaction types can uncover anomalies.

9. newBalanceDest:

- **Explanation:** The balance of the destination account after the transaction.
- **Business Insight:** Similar to `newBalanceOrig`, this allows us to verify transaction integrity: Does `oldBalanceDest + amount = newBalanceDest` (for incoming transactions)? Any significant deviations or suspicious balance changes (e.g., a large `amount` going into an account with a perpetually zero balance) should be investigated further.

10. isFraud:

- **Explanation:** This is the target variable, indicating whether the transaction was actually fraudulent (1) or legitimate (0). This is our ground truth.
- **Business Insight:** This is the most important column for a fraud detection project. We need to calculate the overall fraud rate in the dataset and analyze how `isFraud` correlates with all other features. For example, what percentage of "TRANSFER" transactions are fraudulent compared to "PAYMENT" transactions? This column will be used to train and evaluate any predictive models.

11. isFlaggedFraud:

- **Explanation:** This column indicates whether the transaction was flagged as fraudulent by an existing business rule or system.
- **Business Insight:** This column represents the performance of your current fraud detection system. It's crucial to compare this with `isFraud` to understand the effectiveness of existing rules. We should calculate metrics such as:
 - **Precision:** Out of all transactions flagged as fraudulent by the system, how many were actually fraudulent? ($\text{True Positives} / (\text{True Positives} + \text{False Positives})$)
 - **Recall (or Sensitivity):** Out of all truly fraudulent transactions, how many were correctly 'flagged by the system'? ($\text{True Positives} / (\text{True Positives} + \text{False Negatives})$)
 - **Accuracy:** Overall correct predictions ($(\text{True Positives} + \text{True Negatives}) / \text{Total transactions}$).
- A high number of `isFlaggedFraud=1` where `isFraud=0` indicates a high false positive rate, leading to unnecessary customer friction and operational costs. A high number of `isFlaggedFraud=0` where `isFraud=1` indicates a high false negative rate, meaning significant fraud is being missed. Understanding the specific business rules that trigger `isFlaggedFraud` (e.g., transfers over a certain amount, or specific account types) is essential for evaluating and improving them.

Key Areas for Further Analysis

- **Balance Consistency Checks:** A primary step should be to verify if `oldBalanceOrig - amount = newBalanceOrig` and if `oldBalanceDest + amount = newBalanceDest` (for corresponding transaction types). Any discrepancies here, especially in fraudulent transactions, warrant deep investigation.
- **Zero Balance Anomalies:** Investigate transactions where `oldBalanceDest` is zero but an `amount` is transferred, or where `oldBalanceDest` is zero for a receiving account in a large `TRANSFER` or `CASH_OUT`.
- **Fraudster Modus Operandi:** Analyze the characteristics (transaction type, amount, balance changes) of transactions where `isFraud = 1` to identify common patterns or unique signatures of fraudulent activities.
- **System Performance Evaluation:** A detailed analysis of `isFlaggedFraud` against `isFraud` will provide a baseline for how well the current system performs and highlight areas for improvement in your fraud detection capabilities.

This dataset provides a robust foundation for deep dive analysis into fraud patterns and for building more intelligent fraud detection mechanisms.



AI Insights

🧠 Gemini Business Insights

Here is a business analysis based on the provided dataset and KPIs:

Business Intelligence Analysis: Fraud Detection

1. Key Business Insights

- **Significant Transactional Activity:** The dataset comprises a substantial volume of 6,362,620 total records, representing a massive transaction volume of approximately 1.14 trillion (1,144,392,944,759.77) units of currency. The average transaction value is 179,861.9 units of currency, indicating that transactions involve significant amounts on average.
- **Presence of Fraud:** Out of all transactions, 8,213 cases have been identified as fraudulent.
- **Low Fraud Rate, High Absolute Number:** The overall fraud rate is relatively low at 0.13%. However, the absolute number of fraud cases (8,213) is not negligible and represents a critical area for concern.
- **Predominance of Legitimate Transactions:** The vast majority of transactions, 6,354,407, are legitimate, underscoring the efficiency and normalcy of operations for the bulk of transactions.
- **High Data Quality:** There are no missing values across any of the 11 columns, indicating excellent data integrity for analysis.

2. Important Trends

- **Time-based Analysis Unavailable (from KPIs):** While a 'step' column exists in the dataset which could potentially represent time series data or sequential steps, the provided KPIs are aggregated across all records. Therefore, specific trends over time (e.g., daily, weekly, or by 'step') for fraud rates, transaction volume, or average transaction value cannot be determined from the provided KPIs.
- **Transaction Type/Amount Trends Unavailable:** Trends related to specific transaction types ('type' column) or patterns in transaction amounts for legitimate versus fraudulent activities are not available within the provided KPIs.

3. Risks or Problems

- **Financial Exposure to Fraud:** Although the fraud rate is low (0.13%), the presence of 8,213 fraud cases represents a definite financial risk. The total financial loss due to these fraudulent transactions is not available in the dataset's KPIs, but the average transaction value (179,861.9) suggests that each fraud case could potentially involve a substantial sum.
- **Reputational Damage:** Fraudulent activities can lead to a loss of trust from legitimate users/customers and potential reputational damage, which is not quantifiable with the current metrics but is an inherent business risk.
- **Undetermined Fraud Vectors:** Without a breakdown of fraudulent transactions by 'type', 'amount' range, or other characteristics, the specific methods or common patterns used by fraudsters remain unknown. This makes proactive mitigation strategies difficult to formulate without further analysis.
- **Possible Interpretation:** The presence of an 'isFlaggedFraud' column suggests an internal system or process for flagging suspicious transactions. However, the effectiveness, accuracy, or number of cases flagged by this system are not reflected in the provided KPIs, leaving its value and performance undeterminable from this analysis.

4. Business Recommendations

- **Prioritize In-depth Fraud Analysis:** Conduct a deeper investigation into the 8,213 fraudulent transactions. This should include:
 - Analyzing the 'type' of transactions that are most frequently associated with fraud.
 - Examining the 'amount' characteristics of fraudulent transactions (e.g., minimum, maximum, average amounts for fraud).
 - Investigating patterns related to 'oldbalanceOrig', 'newbalanceOrig', 'oldbalanceDest', and 'newbalanceDest' to identify abnormal balance changes in fraudulent scenarios.
- **Quantify Financial Loss from Fraud:** If the raw data is accessible, calculate the total monetary value lost due to the 8,213 fraudulent transactions. This metric is crucial for understanding the true financial impact and prioritizing fraud prevention efforts.
- **Develop and Refine Fraud Detection Models:** Utilize the detailed information from fraudulent cases to build or enhance machine learning models for proactive fraud detection, focusing on the identified patterns and characteristics.
- **Monitor 'isFlaggedFraud' Performance:** If 'isFlaggedFraud' represents a current detection mechanism, analyze its precision and recall against the 'isFraud' column to understand its effectiveness and identify areas for improvement. (This recommendation is based on the column's existence, not on a KPI showing its performance).
- **Implement Transaction Monitoring by 'Step':** Leverage the 'step' column to analyze fraud trends over time (or sequential steps) if the raw data permits. This can help identify peak periods or specific operational steps where fraud is more prevalent, enabling targeted interventions.



Business Forecasting ↔



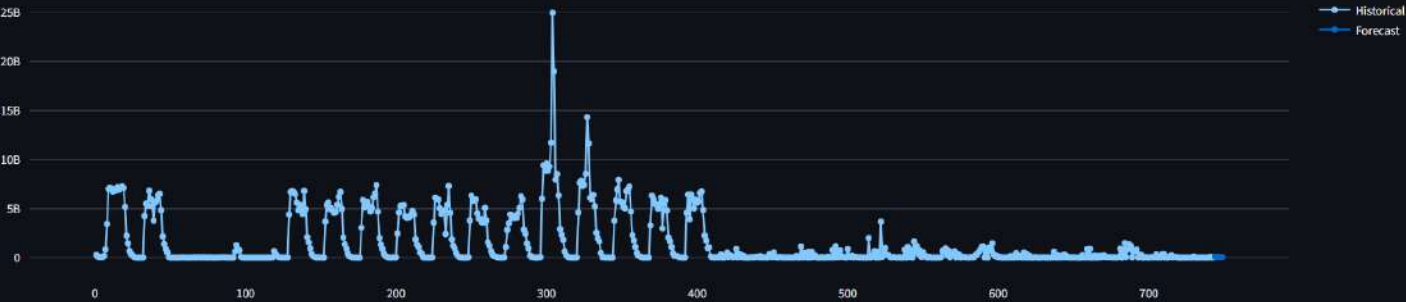
Forecast Capability

Forecast Available: Transaction Volume

Detected Dataset: FRAUD



Transaction Volume Forecast





Forecast Summary

Current Value	Future Prediction	Expected Change
17519825.5	51938348.2	196.45%

Forecast trend: Increasing



AI Forecast Explanation

Here is a short business forecast explanation:

- Forecast Observation** TransactionVolume is predicted to experience a substantial increase. The historical last value was 17,519,825.5, and the future predicted value is 51,938,348.2, representing an expected change of 196.45%.
- Trend Explanation** The forecast indicates an "Increasing" trend for TransactionVolume. This suggests a continuation or acceleration of previous growth patterns within the dataset.
- Assumptions and Limitations** This forecast assumes that the underlying patterns driving TransactionVolume continue as observed historically to generate the predicted future value. External factors that could significantly influence TransactionVolume, such as economic shifts, changes in user base, platform policy updates, or the introduction of new fraud prevention technologies, are not available within the provided forecast information and therefore have not been considered in this explanation.



Future Prediction

	TimeIndex	TransactionVolume	Date
0	743	71677000.9507	744
1	744	67729270.4009	745
2	745	63781539.851	746
3	746	59833809.3012	747
4	747	55886078.7513	748
5	748	51938348.2015	749

Executive Business Report

Detected Dataset Type: FRAUD

Business Overview

Total Records

6362620

Transaction Volume

1144392...

Avg Transaction Value

179861.9

Fraud Cases

8213

Fraud Rate %

0.13

Legit Transactions

6354407

Missing Values

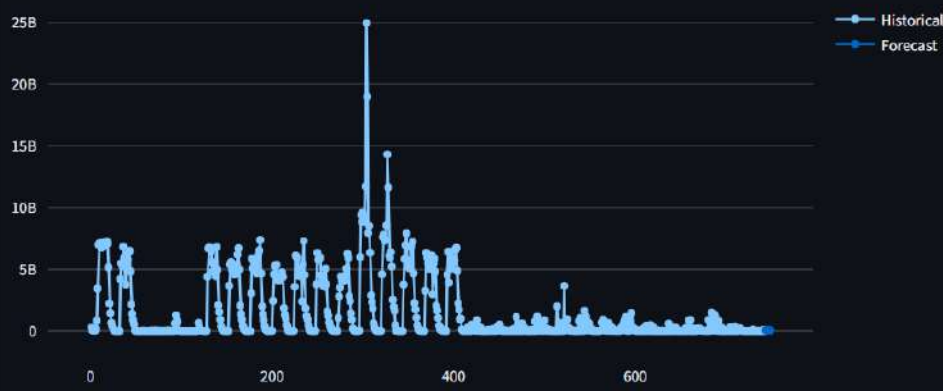
0

Executive Summary

⚠️ AI Insights temporarily unavailable.

Gemini API daily limit has been reached. Please try again later.

Transaction Volume Forecast



Future Prediction

	Date	TransactionVolume
0	744	71677000.9507
1	745	67729270.4009
2	746	63781539.851
3	747	59833809.3012
4	748	55886078.7513
5	749	51938348.2015